

## Planetary Gearbox - 1-Stage Teardown and Repair Analysis

### A. Job information

Customer name

Work order number

Date received

Manufacturer

Model

Serial number

Overall reduction

: 1

Stage 1 ratio

: 1

Oil capacity

Oil grade

Application / driven equipment

Service hours (if known)

Reported failure / customer complaint

### B. As-received condition

☐

Photos taken as received

Notes

☐

External damage, cracks, missing parts

Notes

☐

Nameplate present and legible

Notes

# Ram Gear Gearbox Evaluation

Shafts turn freely by hand: ☐ Yes ☐ No

Notes

☐ Noise or roughness when turned

Notes

☐ Oil leaks at seals / split line / plugs

Notes

☐ Breather condition

Notes

## C. Oil condition

Oil drained - volume recovered

Oil sample ID (lab analysis)

☐ Oil sample taken for lab analysis

Appearance: ☐ Clean ☐ Dark ☐ Milky (water) ☐ Sludge ☐ Metallic particles

Magnetic drain plug debris description

## D. As-found measurements before teardown

Endplay - Input shaft

Endplay - Output shaft

Output rotational backlash (input locked)

Ext. runout - Input shaft

Ext. runout - Output shaft

# Ram Gear Gearbox Evaluation

## E. Teardown checklist

- ☐ Match-mark housing, ring gears, carriers and shafts before disassembly
- ☐ Parts tagged and kept in order, by planetary stage
- ☐ Seals removed and inspected

## As-found shim record

| Shaft  | Drive end (DE) as-found shim (in/mm) | Non-drive end (NDE) as-found shim (in/mm) | Shim pack                        |                                |
|--------|--------------------------------------|-------------------------------------------|----------------------------------|--------------------------------|
| Input  | <input type="text"/>                 | <input type="text"/>                      | <input type="checkbox"/> Replace | <input type="checkbox"/> Reuse |
| Output | <input type="text"/>                 | <input type="text"/>                      | <input type="checkbox"/> Replace | <input type="checkbox"/> Reuse |

## P1. Planetary stage 1

|                                              |                          |                                                      |                      |
|----------------------------------------------|--------------------------|------------------------------------------------------|----------------------|
| Sun gear teeth (Z sun)                       | Ring gear teeth (Z ring) | Planet gear teeth                                    | Number of planets    |
| <input type="text"/>                         | <input type="text"/>     | <input type="text"/>                                 | <input type="text"/> |
| Stage ratio, calculated (1 + Z ring / Z sun) |                          | Stage ratio, measured (input turns per carrier turn) |                      |
| <input type="text"/> : 1                     |                          | <input type="text"/> : 1                             |                      |

## Stage 1 measurements

| Measurement                 | Measured             | Spec                 | OK                       |
|-----------------------------|----------------------|----------------------|--------------------------|
| Sun-planet backlash         | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Planet-ring backlash        | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Sun gear float / axial play | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Carrier endplay             | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Thrust washer thickness     | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |

## Sun gear

Sun gear condition: ☐ No defects ☐ Tooth wear ☐ Pitting / spalling  
☐ Spline wear / fretting ☐ Cracks / breakage

# Ram Gear Gearbox Evaluation

## Stage 1 sun gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Planet gears

### Stage 1 planets (one row per planet)

| Planet   | Gear teeth condition | Pin / shaft wear & fit | Bearing / needle rollers | Thrust washers       | Planet                         |                                 |                                  |
|----------|----------------------|------------------------|--------------------------|----------------------|--------------------------------|---------------------------------|----------------------------------|
| Planet 1 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 2 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 3 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 4 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 5 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |
| Planet 6 | <input type="text"/> | <input type="text"/>   | <input type="text"/>     | <input type="text"/> | <input type="checkbox"/> Reuse | <input type="checkbox"/> Repair | <input type="checkbox"/> Replace |

## Stage 1 planet pins / shafts

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Stage 1 planet bearings / needle rollers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Stage 1 thrust washers

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Ring gear

Ring gear condition: ☐ No defects ☐ Internal teeth wear ☐ Pitting / spalling ☐ Cracks  
☐ Mounting / housing fit worn ☐ Dowels / bolts loose or damaged

# Ram Gear Gearbox Evaluation

## Stage 1 ring gear

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Planet carrier

Carrier condition: ☐ No defects ☐ Cracks ☐ Pin bores worn / oval

☐ Carrier bearing damage ☐ Spline / output coupling wear

## Stage 1 planet carrier

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## Stage 1 carrier bearing

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

## Stage 1 output spline / coupling (stage 1 carrier to output)

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

## F. Input, output, housing and lubrication

### Input shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

### Input seal

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

### Input shaft bearings

☐ Reuse ☐ Repair ☐ Replace

Cone P/N:

Cup P/N:

Findings:

### Output shaft

☐ Reuse ☐ Repair ☐ Replace

Part no.:

Findings:

# Ram Gear Gearbox Evaluation

## Output shaft bearings

☐

Reuse

☐

Repair

☐

Replace

Cone P/N:

Cup P/N:

Findings:

## Output seals

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

## Housing and bearing bores

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

## Lubrication: breather, plugs, sight glass, oil lines

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

## Bearing caps and shims

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

## Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

## Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

## Other:

☐

Reuse

☐

Repair

☐

Replace

Part no.:

Findings:

## G. Failure modes observed

**Gears:**☐

Pitting

☐

Spalling

☐

Scuffing/scoring

☐

Wear

☐

Tooth breakage

☐

Plastic deformation

☐

Corrosion

☐

Misaligned contact pattern

**Bearings:**☐

Spalling/fatigue

☐

Brinelling

☐

Contamination dents

# Ram Gear Gearbox Evaluation

☐ Overheating/discoloration ☐ Corrosion/etching ☐ Electrical fluting

☐ Cage damage ☐ Skidding

**Planetary:** ☐ Planet pin wear ☐ Needle roller damage ☐ Thrust washer wear

☐ Carrier cracks ☐ Spline fretting ☐ Ring gear loose in housing

Location / notes

## H. Dimensional checks

### Bearing bore and journal diameters

| Location                | Bearing bore measured | Bearing bore spec    | Bore OK                  | Journal measured     | Journal spec         | Journal OK               |
|-------------------------|-----------------------|----------------------|--------------------------|----------------------|----------------------|--------------------------|
| Input shaft bearing     | <input type="text"/>  | <input type="text"/> | <input type="checkbox"/> | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Output shaft bearing    | <input type="text"/>  | <input type="text"/> | <input type="checkbox"/> | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |
| Stage 1 carrier bearing | <input type="text"/>  | <input type="text"/> | <input type="checkbox"/> | <input type="text"/> | <input type="text"/> | <input type="checkbox"/> |

Seal journal condition notes

## I. Root cause

☐ Lubrication failure ☐ Contamination ☐ Overload / shock load ☐ Misalignment

☐ Improper installation ☐ Fatigue / end of life ☐ Manufacturing defect

☐ Other

Root cause narrative

## J. Repair summary

Parts to replace

Machining / repair work required

Recommended upgrades or changes

Estimated labor hours

Estimated parts cost

Quote number

**Recommendation:**

☐

Repair

☐

Replace unit

☐

Not economical to repair

## K. Sign-off

### Inspected by

Name

Signature (type name)

Date

### Reviewed / approved by

Name

Signature (type name)

Date

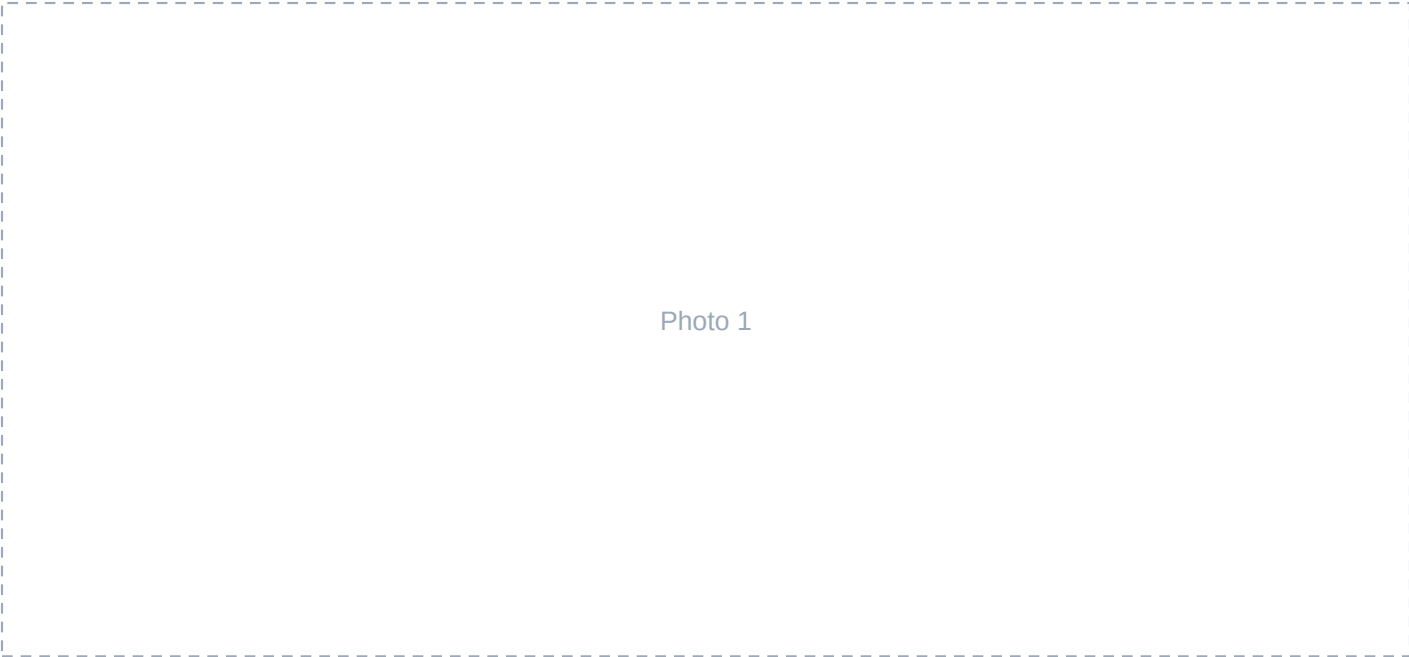
☐

Customer approval received

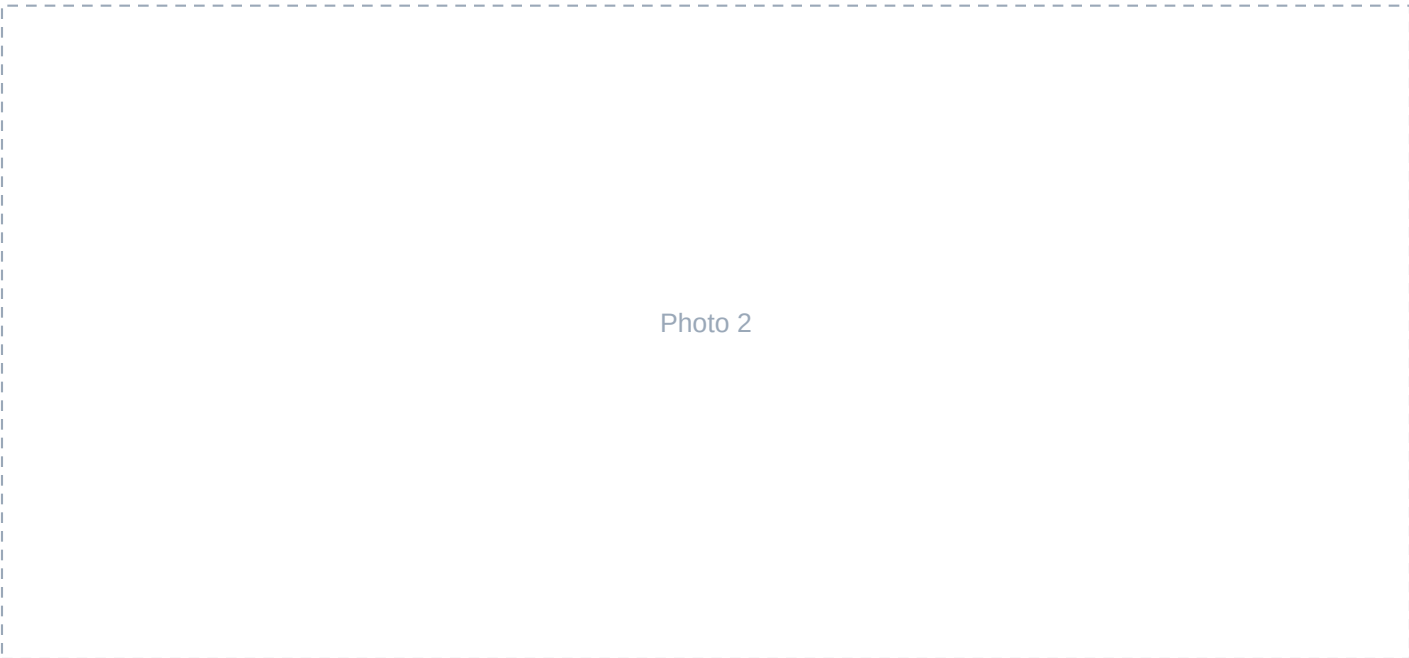
Customer approval date

## L. Teardown photos

Attach teardown photos. Label each photo with a short caption, e.g. As received, Disassembled or a component name. Forms completed in the app print the photos and captions in this section.



Caption: \_\_\_\_\_



Caption: \_\_\_\_\_